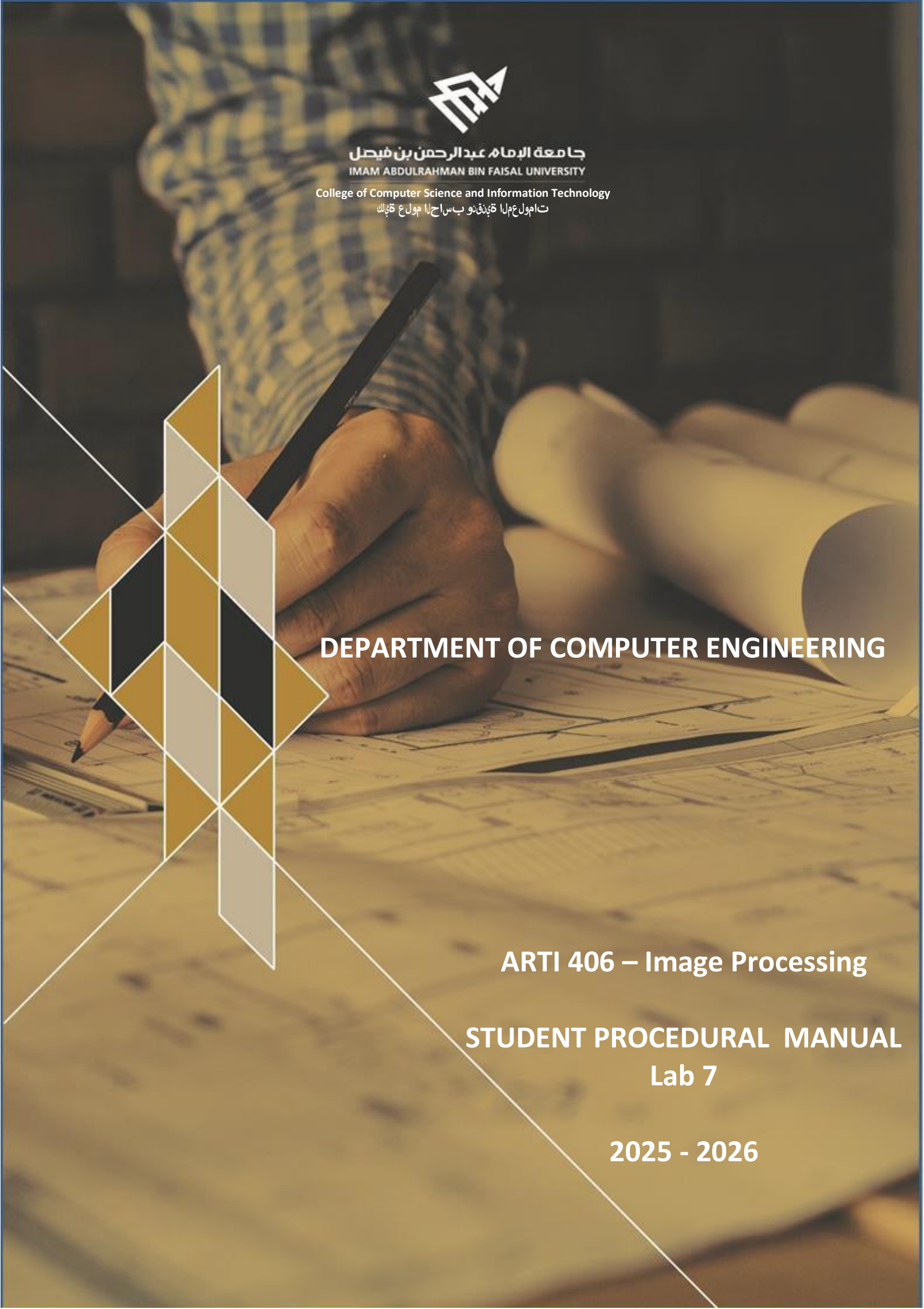

جامعة الإمام عبد الرحمن بن فيصل
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY
College of Computer Science and Information Technology
كلية علوم الحاسب وتقنية المعلومات
DEPARTMENT OF COMPUTER ENGINEERING
ARTI 406 – Image Processing
STUDENT PROCEDURAL MANUAL
Lab 7
2025 - 2026

IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

Lab 7 – Answers: Laplacian Filter & Noise Filters

Task 1: Image Sharpening with the Laplacian Filter

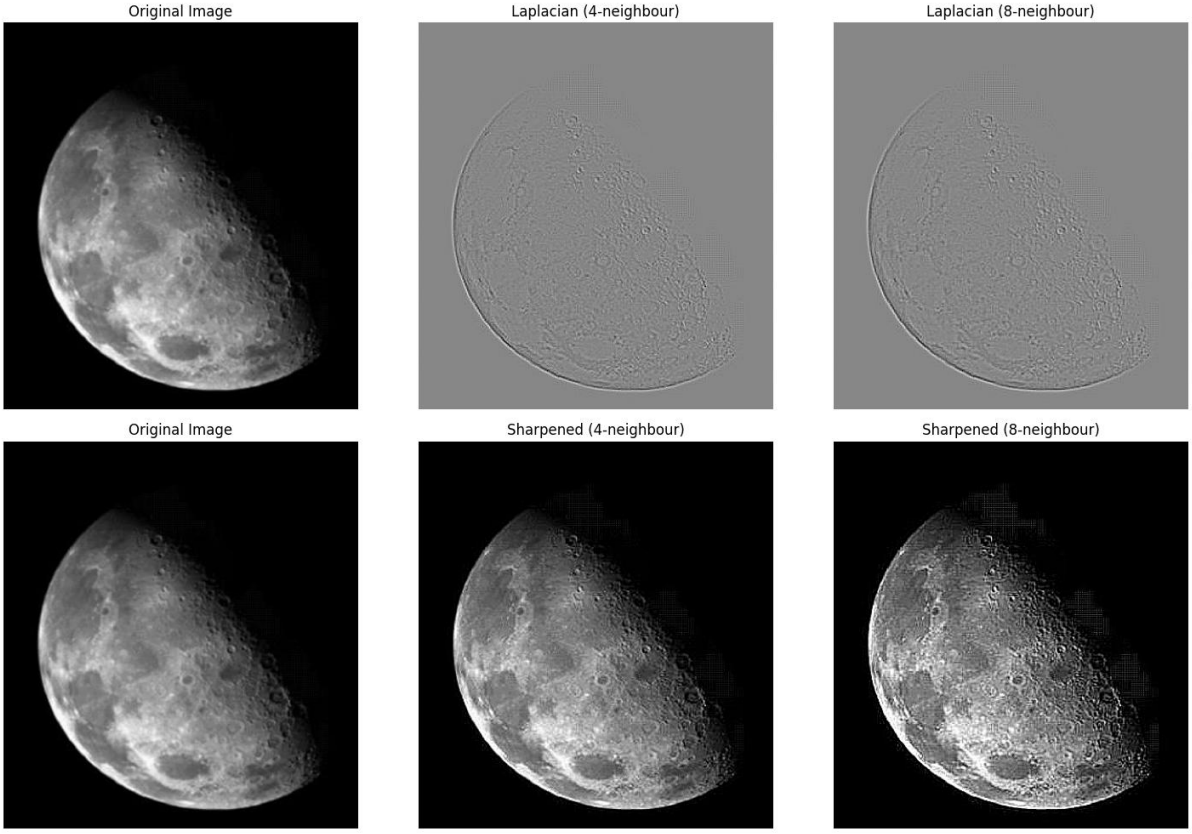


Figure 1: Laplacian images (top) and sharpened images (bottom)

Explanation

The Laplacian filter finds the edges and fine details. Subtracting it from the original image makes the image sharper, so the craters of the moon become clearer.

The 8-neighbour kernel gives stronger sharpening than the 4-neighbour kernel because it also uses the diagonal pixels (sharpness: original 25.7, 4-neighbour 47.4, 8-neighbour 85.5), but it also makes the noise more visible.

Task 2: Removing Noise with Different Filters

I added different types of noise to noise.jpg and tried all the filters. PSNR measures the quality of the result (higher = closer to the original image).

Noise type	Best filter	PSNR (noisy → filtered)	Why
Gaussian	Arithmetic Mean	20.85 → 30.01 dB	Averages the random noise
Uniform	Midpoint	26.88 → 37.14 dB	Noise has a limited range
Exponential	Min	25.68 → 33.45 dB	Noise only makes pixels brighter
Salt & Pepper	Median	18.42 → 51.03 dB	Ignores the black and white dots

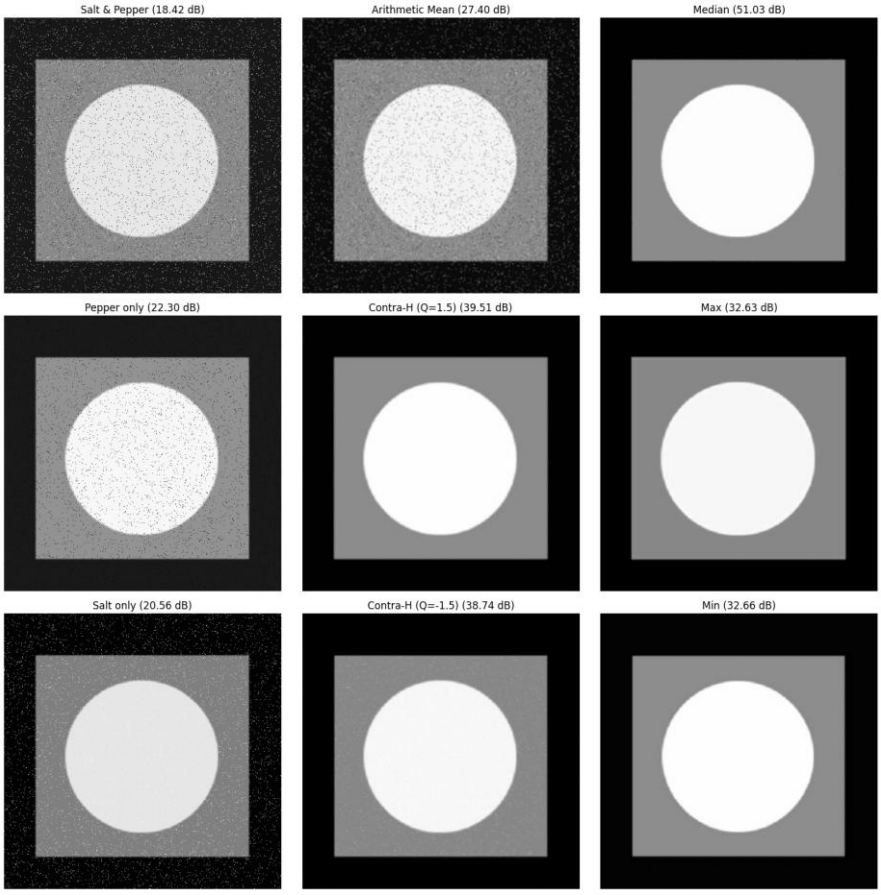


Figure 2: Filters applied to salt & pepper, pepper-only and salt-only noise

Explanation

The median filter is the best for salt & pepper noise. The mean filter only spreads the dots into grey spots.

The contra-harmonic filter with Q > 0 removes pepper (black dots), and with Q < 0 it removes salt (white dots). The max filter removes pepper and the min filter removes salt.

Task 3: Real Salt & Pepper Image

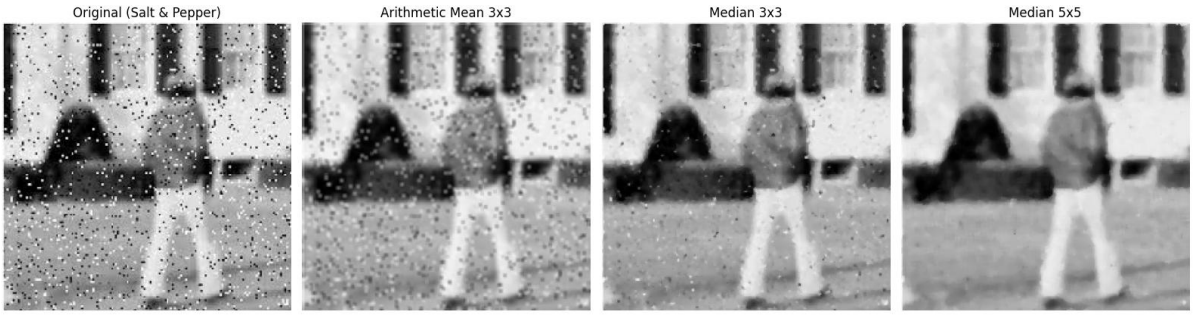


Figure 3: Mean and median filters on salt&Pepper.jpg

The median filter works much better than the mean filter. The 5×5 median removes almost all the noise, but the image becomes more blurred than with the 3×3 median.

Conclusion

The filter must be chosen based on the type of noise. The median filter is the best choice for salt & pepper noise.